

THE HORIZONTAL GEOMETRY OF PRODUCTION NETWORKS

Valerio Dionisi

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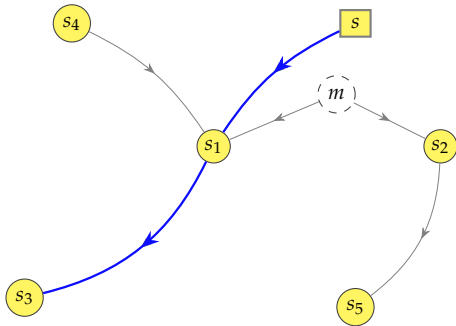
Department of Economics, Management and Statistics

University of Milano-Bicocca

BACKGROUND

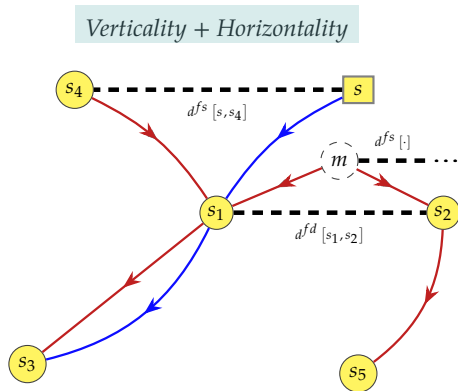
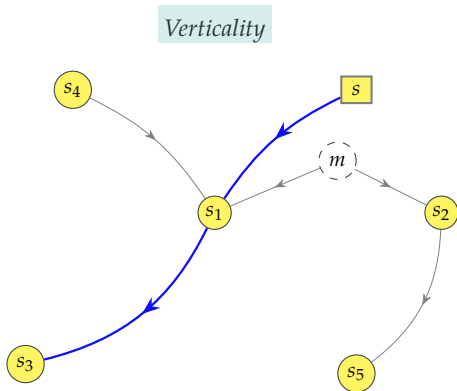
- ▶ Comovement across sectors is an essential feature of business cycles {e.g., Lucas (1977)}
 - driven by independent micro-level shocks, not aggregate {e.g., Gabaix (2011), Foerster *et al.* (2011)}
- ▶ Production networks $\longrightarrow$ role of Input-Output linkages {e.g., Acemoglu *et al.* (2012)}
 - an idiosyncratic shock propagates to others along supply chains $\leftrightarrow$ “cascade” effects;
 - economically dominant sectors generate persistent aggregate fluctuations.
- Channel: *vertical complementarities*, through network trade intensities {e.g., Shea (2002)}
 - *linear economies* $\leftrightarrow$ comovement from even propagation of shocks {most of the literature};
 - *non-linear complementarities in production* $\leftrightarrow$ the even transmission of shocks is reversed, thereby allowing opposite comovement {e.g., Corsetti *et al.* (2008), Baqaee and Farhi (2019)}.

Verticality



THIS PAPER: CENTRAL INTUITION

- ▶ Beside the *vertical* dimension, there exists an *horizontal* geometry of the network
 - 👉 sectors are additionally held together by *demand and supply interdependencies from having in common a similar Input-Output structure*
 - ▶ Horizontality → network “economic” distances between paired sectors
 - *factor input demand* distance, when *buying from* similar upstream sectors;
 - *factor input supply* distance, when *selling to* similar downstream sectors.
 - ▶ These novel horizontal interdependencies add to the vertical transmission of shocks
 - network connections extend beyond trade flows ↔ horizontal mapping
- “who trades with whom” + “who depends on whom”



- Horizontal geometry adds to existing linkages and bounds disconnected nodes

CENTRAL THEORETICAL RESULTS

Question

how does the horizontal transmission impact comovement and aggregate dynamics?

- ① *sectoral comovement*: it depends on the nature of the network “economic” distance
 - ambiguous in *demand*: production shifts away from common upstream sellers;
 - opposite in *supply*: relative price effect from competition for common downstream buyers;
 - minor distance $\leftrightarrow$ stronger horizontal complementarities across nearby sectors.
- ② *aggregate dynamics*: systematic comovement prevent shocks from averaging out
 - sectors sharing upstream suppliers and downstream buyers respond simultaneously;
 - no need for vertical transmission, cascading amplification, and dominant sectors.

EMPIRICAL OUTCOMES

- ▶ Horizontal transmission is about a rebalancing across factors of production
 - *endogenous comovement*: changes in intermediate inputs induce analogous labour changes
 - this matches with my empirical findings on sector-specific variations ▶ Demand ▶ Supply
- ▶ Sectoral employment comovement is well documented {e.g., Cooper and Haltiwanger (1990), Christiano and Fitzgerald (1998)}, but no studies exploit a networked dimension
- ▶ Empirics on U.S. data ratify the theoretical insights on employment comovement
 - ① demand-linked sectors: *muted/mixed*;
 - ② closely supply-linked sectors: *opposite*; if farther apart: *positive*;
 - ③ clear responses for peripheral sectors after a shock in central sectors: *opposite* and *positive*.
- ▶ A data-driven simulation validates the aggregate fluctuations results

RELATED LITERATURE

👉 Advance the understanding of how Input-Output economies work

- ▶ production network theory {e.g., Long and Plosser (1983), Acemoglu *et al.* (2012), Carvalho (2014), Grassi (2017), Oberfield (2018), Carvalho and Grassi (2019), Liu (2019), Baqaee and Farhi (2019), Acemoglu and Azar (2020), vom Lehn and Winberry (2022), La'O and Tahbaz-Salehi (2022), Ghassibe (2024), Huo *et al.* (2025), Qiu *et al.* (2025)}
 - wide range of topics (fiscal and monetary policy, efficiencies and inefficiencies, growth, financial frictions, international contexts) where the network is *vertical* only
 - **Contribution:** first to introduce, explore, and formalize the network's *horizontal geometry*
- ▶ empirics on the functioning of Input-Output linkages {e.g., Conley and Dupor (2003), Acemoglu *et al.* (2015), Barrott and Sauvagnat (2016), Ghassibe (2021), Barattieri and Cacciatore (2023), Barattieri *et al.* (2023)}
 - **Contribution:** first to provide evidence of shocks' transmission due to network distances
- ▶ empirics on comovement of sectoral employment {e.g., Cooper and Haltiwanger (1990), Christiano and Fitzgerald (1998), Cassou and Vazquez (1990), Yedid-Levi (2016), Kim (2020)}
 - **Contribution:** first to take a production network perspective

WHAT'S TO COME

- ① FOUNDATIONAL THEORY, to define network distances and their role on comovement
 - inspect how shocks propagate considering horizontal linkages and input substitutability
- ② AGGREGATE DYNAMICS, to induce aggregate cycles from horizontal complementarities
 - different and independent mechanisms from standard vertical propagation
- ③ EMPIRICS, network effects of sectoral variations to study theoretical results
 - data-driven simulation of aggregate GDP fluctuations;
 - network dimension of nearby sectors' employment changes on sectoral employment.
- ④ POLICY IMPLICATIONS, for future novel perspectives (not today, but in the paper)
 - improving the design of policy interventions through the network's horizontal geometry

FOUNDATIONAL THEORY

GENERAL FRAMEWORK

- ▶ Consider a general production function for sector $\{s, s', s'', \dots, m\} \in \Phi(s)$

$$y(s) = z(s) f^y \left(n(s) ; \{x(s, s')\}_{s' \in \Phi(s)} \right)$$

- ▶ Combination of optimality conditions delivers

$$x(s, s') = f \left(\alpha(s, s'), p(s') ; \alpha(s), n(s), w(s) \right) , \quad \forall s' \in \Phi(s) \quad (1)$$

- $\alpha(s, s')$: intensity of the good from sector- s' in the intermediates bundle of sector- s
- ▶ Then, consider two types of *production network “economic” distances*:
 - ① *factor input demand*, when given sectors are *buying from* the same sector;
 - ② *factor input supply*, when given sectors are *selling to* the same sector.

DISTANCES

- ▶ Exploit the distance's definitions to relate sectoral optimality conditions ▶ Mixed conditions
 - on the input side, sectors buying from the same suppliers rebalance simultaneously their combination of labour and intermediates; ▶ Demand
 - on the output side, sectors selling to the same buyers exhibit relative price changes, and thus variations in downstream demand. ▶ Supply
- ▶ Systematic adjustments scaled by *distance matrices*, $\mathcal{D}^j$ for $j = \{fd, fs\}$ ▶ Interpretation

 **THEOREM 1** formalizes their characteristics, such as: ▶ Theorem 1

- *non-negative* and *full*: all sectors are horizontally connected, even if not trading vertically;
- a unique Input-Output structure has different network distance relations;
- conditions under which distances are uniquely determined (demand = supply). ▶ Corollary 1

DISTANCE-LEAD COMOVEMENT

- ▶ So far intermediate inputs generally function as substitutes
 - non-unitary substitution elasticity alters propagation {e.g., [Atalay \(2017\)](#), [Baqaei and Farhi \(2019\)](#)}
- ▶ Reduced-form intuition for opposite comovement under distances

► Propositions

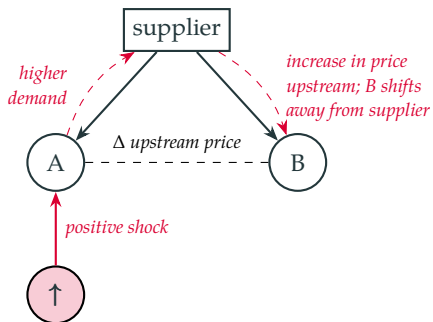
$$\text{demand-driven comovement : } \underbrace{\frac{\partial \log x(s, s^*)}{\partial \log x(s', s^*)}}_{<0} \propto -\sigma [1 - e(s, s^*)] \underbrace{\frac{\partial \log p(s^*)}{\partial \log x(s', s^*)}}_{>0}$$

$$\text{supply-driven comovement : } \underbrace{\frac{\partial \log x(s^*, s)}{\partial \log x(s^*, s')}}_{<0} \propto \sigma e(s^*, s') \underbrace{\frac{\partial \log p(s')}{\partial \log x(s^*, s')}}_{<0}$$

👉 Substitution elasticities matter in magnitude, not in the direction of comovement

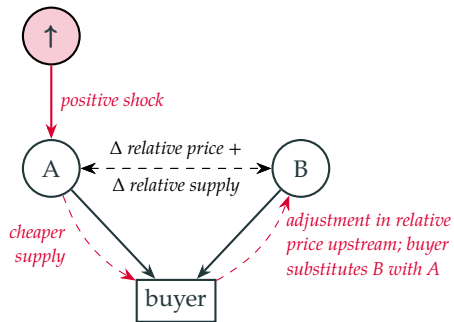
GRAPHING THE ECONOMICS OF REDUCED FORMS

Demand distance: upstream pass-through



downstream shifting away from upstream input
 $\Rightarrow$ vertical price effect + horizontal ambiguity downstream

Supply distance: downstream pass-through



upstream inputs used jointly downstream
 $\Rightarrow$ demand reallocation and clear horizontal transmission

MAIN THEORETICAL TAKEAWAYS ON COMOVEMENT

- ① the *distance network* connects sectors not just vertically, but also horizontally:
 - introducing the *network “economic” distance* allows to consider also how sectors are related through *horizontal complementarities* in common demand and supply linkages ...
... even if they are neither directly nor indirectly connected in the production network.
- ② *factor input demand* distance relates to changes in relative factors of production
 - substitution patterns across common upstream suppliers
 - *hard to interpret*: vertical propagation + horizontal complementarities
- ③ *factor input supply* distance relates to changes in marginal revenues from trade
 - competition among sectors whose inputs are complementary in downstream buyers
 - *co-movement is transparent*: horizontal relative price effects induce opposite patterns

AGGREGATE DYNAMICS

INSIGHTS FOR AGGREGATE BUSINESS CYCLE

- ▶ Equilibrium adjustments are central to understand the origins of aggregate cycles
- ▶ Fluctuations in aggregate real GDP in the model {e.g., vom Lehn and Winberry (2022)}:

$$d \log Y = \sum_{s \in \Phi(s)} \left(\lambda(s) d \log z(s) + \nu(s) d \log n(s) \right)$$

that is, under the network's horizontal geometry,

$$d \log Y = \sum_{s \in \Phi(s)} \lambda(s) d \log z(s) + \sum_{s \in \Phi(s)} \begin{cases} \nu(s) \Delta \text{comovement} & \text{under major distance} \\ \nu(s) \Delta \text{reallocation} & \text{under minor distance} \end{cases}$$

👉 different network distance values generate different fluctuations in aggregate output

EMPLOYMENT COMOVEMENT

- ▶ How does sectoral employment fluctuate? $\longleftrightarrow$ *vertical* + *horizontal* dimensions

- ▶ Overall network effects on sectoral comovement of employment:

► Components

$$d \log \mathbf{N} = \Theta \left\{ d \log C + d \log \mathbf{S} + d \log \mathbf{V} + d \log \mathbf{D}(j) \right\}, \quad \text{for } j = \{fd, fs\}$$

with $\mathbf{D}(fd) = \mathbf{D}^{fd} \left[d \log \mathbf{N} - \left(\mathbf{L}'_{\Phi(s)} d \log \mathbf{N} \right) \mathbf{1} \right]$ and $\mathbf{D}(fs) = \mathbf{D}^{fs} \left[d \log \mathbf{P} - \left(\mathbf{L}'_{\Phi(s)} d \log \mathbf{P} \right) \mathbf{1} \right]$
denoting the difference between the change in all other sectors and the sector's own change

- ▶ Consistent interpretation:
 - *ambiguous* comovement in demand, as employment changes enter in both $\mathbf{V}$ and $\mathbf{D}^{fd}$;
 - *opposite* comovement in supply, due to sectoral relative-price effects in $\mathbf{D}^{fs}$;
 - vertical propagation prevails under low proximity, leading to *positive* comovement.

TO WHAT EXTENT HORIZONTALITY CONTRIBUTES TO AGGREGATE DYNAMICS?

 **THEOREM 4** → network distances alone generate persistent aggregate fluctuations

$$\liminf_{m \rightarrow \infty} \text{var}(Y) > 0$$

- even absent economically dominant sectors and vertical production chains;
- they ensure persistence, while granularity magnifies aggregate volatility.

► Theorem 4

► Corollary 3

- While *vertical propagation* inherently amplifies idiosyncratic shocks ...
 - as shocks feed into sectors along production chains {e.g., Long and Plosser (1983)}
 - thereby generating cascading effects and aggregate responses {e.g., Acemoglu et al. (2012)}
- ... *horizontal propagation* induces comovement and not recursive feedback loops
 - sectors sharing suppliers or buyers respond simultaneously to horizontal shocks
 - no cascades, but systematic correlation that survives aggregation

VERTICAL AND HORIZONTAL NON-DEPENDENCE

- Finally, a dimensional separation is established

 **THEOREM 2:** the horizontal geometry remains unweighted by vertical intensities

- the distance matrix is not merely a transformation of the Input-Output matrix, but a complementary object that reveals otherwise hidden propagation channels

► Theorem 2

 **THEOREM 3:** the horizontal geometry is increasingly important as the network expands

$$\left\| \mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m \right\|_{\Delta} / \left\| \mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m \right\|_{\Delta} \xrightarrow{m \rightarrow \infty} 0$$

- for larger production networks (e.g., firm-to-firm network), vertical buyer-supplier chains become too sparse, while horizontal complementarities grow denser;
- even when vertical chains expand endogenously as the network grows in size.

► Theorem 3

► Corollary 2

EMPIRICS



- ▶ From the United States' Bureau of Economic Analysis (BEA) I collect information, for 3-digit U.S. 2017 NAICS sectors, on *employment* and on:
 - ▶ I-O data
 - ▶ I-O structure
- inter-sectoral linkages exploiting Input-Output (I-O) matrices:
 - ★ share of commodity production (*market share, after redefinitions* tables);
 - ★ I-O weights in producer values (*commodity-by-commodity total direct requirements* tables).
- disregard small I-O transactions {e.g., [Acemoglu et al. \(2012\)](#), [Carvalho \(2014\)](#)} to build distances
 - ★ no link if sector-s's total input sales/purchases are less than 1%
- ▶ *Unit*: 65 private sectors at 3-digit U.S. 2017 NAICS
- ▶ *Period*: 1998-2022
- ▶ *Network structure* at 2007

robust at different years

SETTING THE STAGE

- Degree of separation between sectors within the network (low value, low distance)

- Entries of distance matrices are *Euclidean distances*, so that $d^j[s, s'] \in [0, 1]$ ► Intensive margin

$$d^{fd}[s, s'] = \sqrt{[\alpha(s, s) - \alpha(s', s)]^2 + [\alpha(s, s') - \alpha(s', s')]^2 + \dots + [\alpha(s, m) - \alpha(s', m)]^2}$$

from the *directed* Input-Output network, and not its Leontief transformation

- By Theorem 2, consider *extensive margins*, $d(s, s') = \{1, 2, \dots, d_{max}\}$ ► Graph ► Characteristics

★ computed as a *shortest path problem* (how many steps to reach other nodes) ► Algorithm

- Before the empirics, I simulate aggregate GDP responses to sectoral AR(1) shocks

$$Y_t = \boldsymbol{\lambda}' \mathbf{y}_t \quad \longleftrightarrow \quad Y_t = \boldsymbol{\lambda}' \mathcal{P}^{(\mathcal{V}, \mathcal{D})} (\rho_t \boldsymbol{\epsilon}_0)$$

exploiting the network's vertical (Input-Output chains) and horizontal (shared buyers/suppliers) propagation structures, relying on observed matrices and data

AGGREGATE FLUCTUATIONS

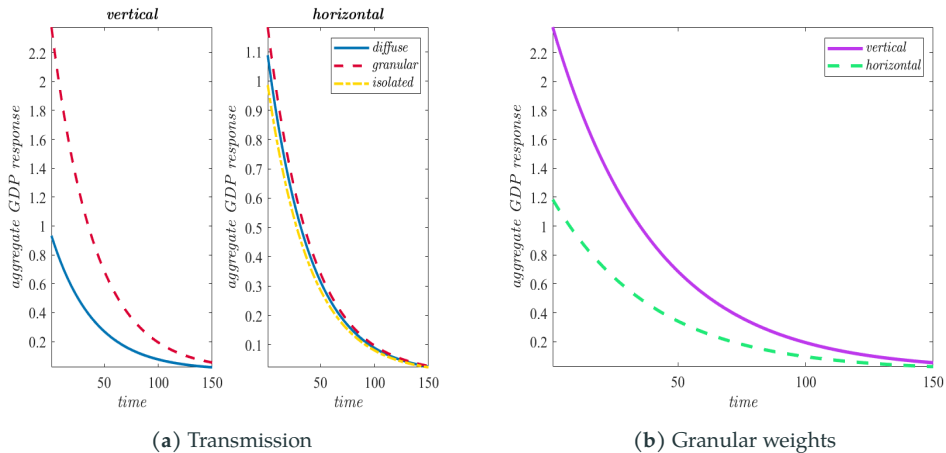


Figure 3: Aggregate fluctuations

EMPLOYMENT CHANGES IN THE NETWORK

- ① IDENTIFICATION, to extract exogenous dynamics {as in Barattieri and Cacciatore (2023)}

$$\begin{aligned}
 n_t(s) = & \underbrace{\beta_y \dot{y}_t(s)}_{\text{output growth}} + \underbrace{\sum_d \beta_{y(d)} \dot{y}_t(\Phi_s, d)}_{\text{output growth, network distance}} + \underbrace{\sum_k \beta_{y(j)} \dot{y}_t(\Phi_s, k)}_{\text{output growth, upstream and downstream}} \\
 & + \underbrace{\beta_{z(s)} Z_t(s) + \beta_{\dot{z}(s)} \dot{Z}_t(s)}_{\text{sector-level controls}} + \underbrace{\beta_z Z_t + \beta_{\dot{z}} \dot{Z}_t}_{\text{aggregate controls}} + \underbrace{\psi(s)}_{\text{sector Fe}} + u_t^n(s)
 \end{aligned}$$

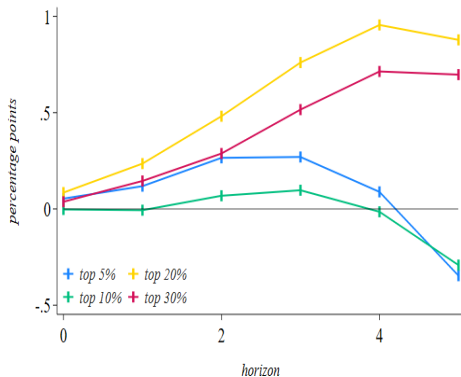
- estimated residuals $(\hat{u}_t^n(s))$ identify the residual variation in sectoral employment

- ② LOCAL PROJECTIONS, for short-run dynamics response {as in Barattieri and Cacciatore (2023)}

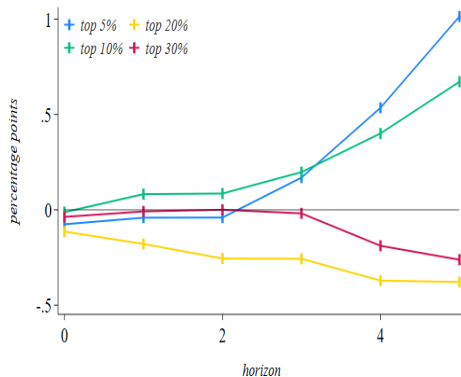
$$\dot{n}_{t+\hbar}^{cum}(s) = \beta_{\hat{n}(d), \hbar} \sum_{s' \neq s} \hat{u}_t^n(s', d) + \underbrace{\psi(t+\hbar) + \psi_{\hbar}(s)}_{\text{horizon and sector Fe}} + v_{t+\hbar}^n(s, d)$$

EMPLOYMENT EFFECTS OF MOST CONNECTED SECTORS, DISTANCE

factor input demand and directed network



(a) $d = 1$, directed

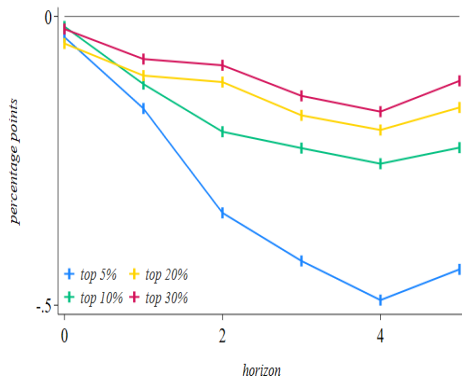


(b) $d = 2$, directed

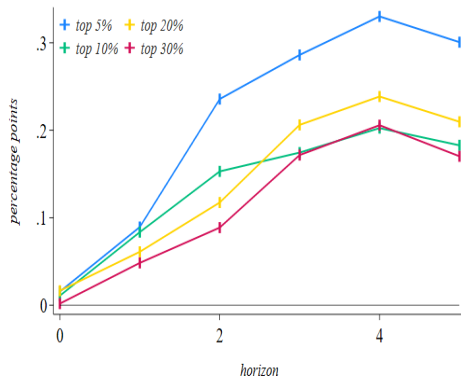
Figure 4: Demand linkages and comovement in interlinked sectors

EMPLOYMENT EFFECTS OF MOST CONNECTED SECTORS, DISTANCE

factor input supply and directed network



(a) $d = 1$, directed

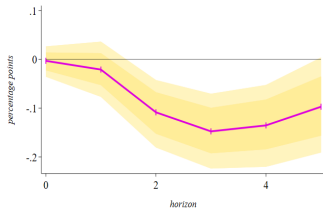


(b) $d = 2$, directed

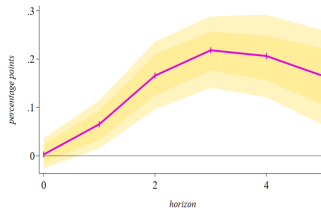
Figure 5: Supply linkages and comovement in interlinked sectors

EMPLOYMENT EFFECTS IN MOST PERIPHERAL SECTORS, DISTANCE

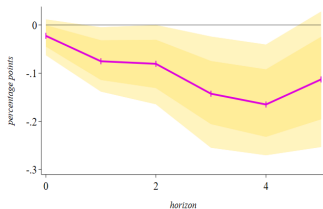
factor input demand, factor input supply and directed network



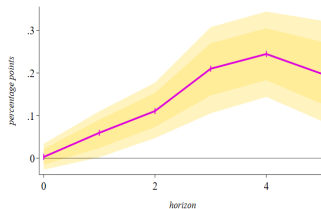
(a) $d = 1$, demand



(b) $d = 2$, demand



(c) $d = 1$, supply



(d) $d = 2$, supply

POLICY IMPLICATIONS

- 👉 Horizontal complementarities are crucial for the transmission of sectoral variations
- ▶ *Fiscal policy*, typically generating upstream transmission { e.g., [Barattieri et al. \(2025\)](#) };
 - it works by expanding sector-specific demand, and thus the supply of others;
 - horizontal geometry is at the core of the effects from government spending;
 - targeted government spending, fiscal multiplier, tariff policy?
- ▶ *Industrial policy*, characterised by two crucial results:
 - market imperfections accumulate through backward demand linkages { [Liu \(2019\)](#) }, and positive effects on buyers of targeted sectors { [Lane \(2025\)](#) };
 - how do interventions affect untargeted sectors exhibiting similar Input-Output structure?
- ▶ *Transmission mechanism*, role of (consumption or production) complementarities
 - asymmetric supply shock is a demand-like shock at the aggregate level { [Guerrieri et al. \(2022\)](#) };
 - horizontal complementarities may clearly, easily and linearly identify demand and supply forces within the network that guide such Keynesian transmission.

- ▶ On the modelling side, expanded framework for:
 - global production networks, extending beyond vertical supply chains {e.g., [Huo et al. \(2025\)](#)};
 - firm-to-firm networks, showing high sparsity {e.g., [Bohem et al. \(2019\)](#), [Carvalho and Grassi \(2019\)](#)};
 - endogenous network formation and inputs specificity {e.g., [Carvalho and Voigtländer \(2015\)](#)};
 - complementarity and sectoral shocks to aggregate outcomes {e.g., [Atalay \(2017\)](#)}.
- ▶ Horizontal geometry is well suited for broader class of economic interdependencies, including investment {e.g., [vom Lehn and Winberry \(2022\)](#)} and financial {e.g., [Acemoglu et al. \(2015\)](#), [Huremovic et al. \(2025\)](#)} networks

CONCLUSION

TAKE-AWAY REMARKS

- ✱ *Horizontal dimension* of a network complements the vertical dimension
 - 👉 it is not the mere existence of Input-Output connections, but **their demand-supply geometry that critically shapes the transmission of micro-originated shocks**
- ▶ Horizontal geometry captured by *network “economic” distances* between sectors
 - ① demand-based distances work through production inputs dynamics, while supply-based distances induce opposite comovement due to relative prices and revenue reallocation;
 - ② persistent aggregate effects even without vertical cascading effects and central sectors;
 - powerful and independent refinement to prevailing network analysis.
- ▶ Comprehensive framework to build distance matrices and form theoretical insights
 - test and corroborate theory-based predictions on sectoral U.S. Input-Output data;
 - suitable for analysis involving the network structure and to design policy prescriptions.

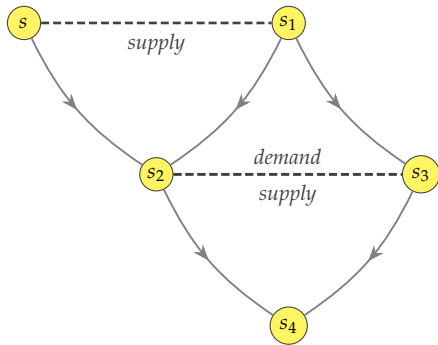
APPENDIX



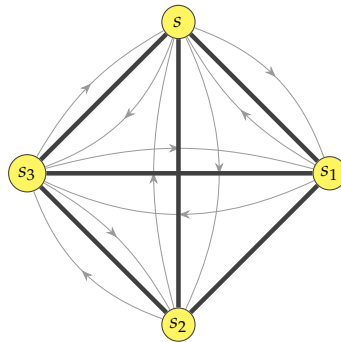
ASSUMPTION 1 (Production technology requirements)

As main characteristics of a production function: (i) it has constant returns to scale in either $x(s, s' \in \Phi(s))$ and $n(s)$, so that production inputs shares sum to one, $\alpha(s) + \sum_{s'} \alpha(s, s') = 1$; (ii) it is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in $z(s)$, $x(s, s' \in \Phi(s))$, and $n(s)$; (iii) the case $f^y(0, \cdot)$ is ruled out as labour input is essential to production; (iv) at least two elements of $x(s, s' \in \Phi(s))$ have to be positive, thus $f^y(\cdot, 0)$ cannot exist; consequentially, (v) part of the sectoral output is directly produced by the sector, $x(s, s) > 0, \forall s \in \Phi(s)$, that is a production plan with a bundle of intermediate inputs as $f^y(\cdot, x(s, s' \in \Phi_s))$ is precluded.

- Note that any functional form chosen for the production function must satisfy the above regularity conditions



(a) Distance definition



(b) Distance network

Figure A.1: Horizontal Input-Output geometries

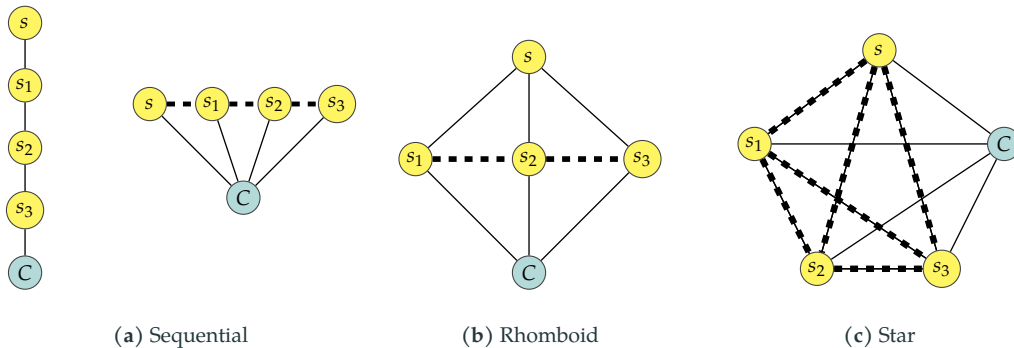


Figure A.2: Benchmark economies

- Solving the system for *factor input demand* network distance

$$\begin{cases} x(s, s^*) = \alpha(s, s^*) w(s) n(s) \alpha(s)^{-1} \frac{1}{p(s^*)} \\ x(s', s^*) = \alpha(s', s^*) w(s') n(s') \alpha(s')^{-1} \frac{1}{p(s^*)} \end{cases}$$

would deliver the condition $\frac{n(s')}{x(s', s^*)} = \frac{\alpha(s, s^*)}{\alpha(s', s^*)} \frac{n(s)}{x(s, s^*)}$.

- Solving the system for *factor input supply* network distance

$$\begin{cases} x(s^*, s) = \alpha(s^*, s) w(s^*) n(s^*) \alpha(s^*)^{-1} \frac{1}{p(s)} \\ x(s^*, s') = \alpha(s^*, s') w(s^*) n(s^*) \alpha(s^*)^{-1} \frac{1}{p(s')} \end{cases}$$

would deliver the condition $\frac{x(s^*, s)}{x(s^*, s')} = \frac{\alpha(s^*, s)}{\alpha(s^*, s')} \frac{p(s')}{p(s)}$.

- ▶ Substituting out eq. (1) for two sectors, say $\{s, s'\}$, actually *buying* their intermediate inputs from the same sector, say s^* , then total *log*-differentiation leads to

$$\underbrace{\left[\gamma_{s'}^B d \log n(s') - \delta_{s'}^B d \log x(s', s^*) \right]}_{\Delta \text{ relative input share of sector-} s'} = \underbrace{\frac{\alpha(s, s^*)}{\alpha(s', s^*)}}_{d_{s^* \rightarrow [s, s']}} \underbrace{\left[\gamma_s^B d \log n(s) - \delta_s^B d \log x(s, s^*) \right]}_{\Delta \text{ relative input share of sector-} s}$$

- ▶ Iteratively, the above condition can be written for all sectors from which the pair is buying. Stacked the outcome for all sectors, then

▶ Lemmas

$$\mathcal{F}_{m \times m} = \mathcal{D}_{m \times m}^{fd} \quad (\text{A.1})$$

- 👉 In other words, the comovement of inputs of production across any pair of sectors is determined by their distance in the production network.

- ▶ Substituting out eq. (1) for two sectors, say $\{s, s'\}$, actually *selling* their intermediate inputs to the same sector, say s^* , then total *log*-differentiation leads to

$$\underbrace{\left[\gamma_{s'}^Q d \log x(s^*, s) - \delta_{s'}^Q d \log x(s^*, s') \right]}_{\Delta \text{ relative quantity of intermediate inputs}} = \underbrace{\frac{\alpha(s^*, s)}{\alpha(s^*, s')}}_{d_{\rightarrow s^*}[s, s']} \underbrace{\left[\gamma_s^P d \log p(s') - \delta_s^P d \log p(s) \right]}_{\Delta \text{ relative price of intermediate inputs}}$$

- ▶ Iteratively, the above condition can be written for all sectors to which the pair is selling. Stacked the outcome for all sectors, then

► Lemmas

$$\mathcal{R}_{m \times m} = \mathcal{D}_{m \times m}^{fs} \quad (\text{A.2})$$

- 👉 In other words, the comovement across (marginal) revenues of production across any pair of sectors is determined by their distance in the production network.

LEMMA 1 (Factor input demand)

Changes in the ratio of production inputs' quantities across any pair of sectors buying from the same sector(s) is driven by their ratio of intermediate intensities with whom the pair is purchasing from: $\mathcal{F} = \mathcal{D}^{fd}$.

LEMMA 2 (Factor input supply)

Changes in the ratio of intermediate input profits across any pair of sectors selling to the same sector(s) is driven by their ratio of intermediate intensities with whom the pair is trading to: $\mathcal{R} = \mathcal{D}^{fs}$.

► Go back, demand

► Go back, supply

- ▶ On the theoretical side, each cell identifies “network proximity”
 - the higher $d[s, s']$, the larger the transmission $\longleftrightarrow$ stronger horizontal complementarity
- ▶ Once taken to the data, each cell measures “network distance”, computed as:
 - *Euclidean distance*, $d^{fd}[s, s'] = \sqrt{[\alpha(s, s) - \alpha(s', s)]^2 + \dots + [\alpha(s, m) - \alpha(s', m)]^2}$;
 - each matrix is *symmetric* outside the main diagonal, with $diag(\mathcal{D}^j) = 0$.
- ▶ These are the *intensive margin*, with values inside $[0, \infty)$. To account for extensive margin(discrete) values, I employ the *shortest path algorithm* (how many steps are necessary for a sector to reach another one)
- ▶ Horizontal dimension is not weighted by vertical dimension (Theorem 2)

▶ Matrix configuration

▶ Theorem 2

▶ Go back

- The *network demand-distance matrix* is defined as

$$\mathcal{D}_{m \times m}^{fd} = \begin{bmatrix} \left(\frac{\alpha(s,s)}{\alpha(s,s)}, \dots, \frac{\alpha(s,m)}{\alpha(s,m)} \right) & \left(\frac{\alpha(s,s)}{\alpha(s',s)}, \dots, \frac{\alpha(s,m)}{\alpha(s',m)} \right) & \cdots & \left(\frac{\alpha(s,s)}{\alpha(m,s)}, \dots, \frac{\alpha(s,m)}{\alpha(m,m)} \right) \\ \left(\frac{\alpha(s',s)}{\alpha(s,s)}, \dots, \frac{\alpha(s',m)}{\alpha(s,m)} \right) & \left(\frac{\alpha(s',s)}{\alpha(s',s)}, \dots, \frac{\alpha(s',m)}{\alpha(s',m)} \right) & \cdots & \left(\frac{\alpha(s',s)}{\alpha(m,s)}, \dots, \frac{\alpha(s',m)}{\alpha(m,m)} \right) \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{\alpha(m,s)}{\alpha(s,s)}, \dots, \frac{\alpha(m,m)}{\alpha(s,m)} \right) & \left(\frac{\alpha(m,s)}{\alpha(s',s)}, \dots, \frac{\alpha(m,m)}{\alpha(s',m)} \right) & \cdots & \left(\frac{\alpha(m,s)}{\alpha(m,s)}, \dots, \frac{\alpha(m,m)}{\alpha(m,m)} \right) \end{bmatrix} = [d^{fd}(s, s')]$$

- Under due adjustments, analogous is the supply-distance matrix: $\mathcal{D}_{m \times m}^{fs} = [d^{fs}(s, s')]$

THEOREM 1 (On the configuration of network “economic” distances)

Let the set of sectors being $\Phi(s)$, and $\alpha^j(\cdot)$, for $j = \{fd, fs, \}$, the normalized Input-Output fixed coefficients. Then, demand- and supply-based distance matrices satisfy: (i) for any $s, s' \in \Phi(s)$ there exists a common neighbour- s^* , so that both $\mathbf{D}^{fd}$ and $\mathbf{D}^{fs}$ are non-negative and full; (ii) a given Input-Output structure uniquely determines $\mathbf{D}^{fd}$ and $\mathbf{D}^{fs}$, which differ entrywise unless the corresponding bilateral trade with the common neighbour coincides; and (iii) if $\alpha^j(\cdot)$'s are non-symmetric, then all entries of $\mathbf{D}^{fd}$ and $\mathbf{D}^{fs}$ differ; conversely, if $m \geq 4$, then both share at least one identical entry whenever at least $m - 2$ sectors trade symmetrically with a common sector and at least one additional symmetric trade relation elsewhere in the network exists.

COROLLARY 1 (Coinciding distances)

For $m \geq 4$, if a common sector trades symmetrically with $S = m - 2$ others, then $\frac{S(S-1)}{2}$ sectors exhibit symmetric distances.

PROPOSITION 1 (Demand-driven comovement)

In the single-input case, comovement in demand implies $\delta_{s^* \rightarrow [s, s']} \propto -\sigma \tau^{fd}$, determined by substitution elasticity σ and upstream pass-through τ^{fd} . In the multi-input CES case, then

$$\delta_{s^* \rightarrow [s, s']} \propto -\sigma [1 - e(s, s^*)] \tau^{fd}$$

with relative intermediate inputs changes from common supplier, $\delta_{s^* \rightarrow [s, s']} = \frac{\partial \log x(s, s^*)}{\partial \log x(s', s^*)}$, jointly determined by substitution elasticity, downstream sectoral expenditure share, $e(\cdot)$, and upstream pass-through. Opposite comovement occurs if $\tau^{fd} > 0$.

PROPOSITION 2 (Supply-driven comovement)

In the single-input case, no condition exists for comovement in supply due to the absence of competition for downstream markets. Differently, in the multi-input CES case,

$$\delta_{\rightarrow s^*} [s, s'] \propto \sigma e(s^*, s') \tau^{fs}$$

with relative intermediate inputs changes to common buyer, $\delta_{\rightarrow s^*} [s, s'] = \frac{\partial \log x(s^*, s)}{\partial \log x(s^*, s')}$, jointly determined by substitution elasticity, σ , sectoral revenues for upstream supplier, $e(\cdot)$, and downstream pass-through. Opposite comovement occurs if $\tau^{fs} < 0$.

- ▶ Intuition beyond the horizontal transmission:
 - *demand distance*: shocks trigger downstream buyer to shift away from upstream suppliers
 - ★ vertical and horizontal effects get mixed, making comovement ambiguous to interpret
 - *supply distance*: upstream shocks pass downstream, where inputs are used together
 - ★ horizontal propagation among suppliers is direct, mechanical, and unambiguous
- ▶ Core contrast:
 - *demand distance*: mere variation in upstream price (no true horizontal complementarities);
 - *supply distance*: substitution of upstream inputs through downstream demand reshuffling.
- ▶ Downstream complementarities (*i.e.*, supply distances) matter the most to generate opposite comovement {in line with [Atalay \(2017\)](#) and [Baqaei and Farhi \(2019\)](#)}

► *Demand-based network distances*

- capture how easily a downstream buyer can switch among suppliers, without creating “true” horizontal linkages $\longleftrightarrow$ *upstream pass-through effect*;
- reflect substitution patterns across buyers sharing common upstream suppliers;
- ambiguous propagation effects, as they lack genuine complementarities among buyers.

► *Supply-based network distances*

- generate genuine horizontal complementarities among sectors whose outputs are jointly used by common downstream buyers $\longleftrightarrow$ *downstream pass-through effect*.
- shocks to one supplier propagate horizontally as other suppliers adjust production due to downstream complementarities;
- strong comovement effects, consistent with non-linear network theories emphasizing complementarities in intermediate inputs. {e.g., [Atalay \(2017\)](#), [Baqae and Farhi \(2019\)](#)}

- (a) $\mathbf{N} = [n(s), n(s'), \dots, n(m)]$ identifies an $m \times 1$ vector of sectoral employment levels;
- (b) $\Theta = \frac{1}{1+\phi+\alpha}$ is a compound of structural parameters, and $\underset{m \times 1}{\boldsymbol{\alpha}} = [\alpha(s), \dots, \alpha(m)]$;
- (c) C is aggregate consumption;
- (d) $d \log \mathcal{S} = d \log \mathbf{z} - d \log \mathbf{C}$ identifies the $m \times 1$ vectors for sector-specific changes in productivity and consumption, respectively;
- (e) $d \log \mathcal{V} = \mathbf{H}(\Psi_{s,s'}) [d \log \mathbf{z} + \boldsymbol{\alpha} d \log \mathbf{N}] + \boldsymbol{\mathcal{E}} d \log \mathbf{X}$ identifies the production network effect of other sectors' changes in productivities, employment levels, and intermediate inputs usage, where:
 - $\mathbf{H}(\Psi_{s,s'})$ is the Input-Output matrix, comprising the weight of each sector on other sectors, whose entries are set to 0 whenever $s = s'$;
 - $\boldsymbol{\mathcal{E}} = \mathbf{H}(\Psi_{s,s'})' \mathbf{H}(\Psi_{s',s})$ is a compounded network effect, made of the inner product of the Input-Output matrix, influencing the elements of $\mathbf{X}$.

THEOREM 4: AGGREGATE FLUCTUATIONS

A

THEOREM 4 (On aggregate cycles)

Consider an Input-Output economy of m sectors. Let $\lambda(s)$ denote sector- s 's Domar weight, and assume no sector is dominant, in the sense that $\lambda(s) \asymp \frac{1}{m}$, $\forall s \in \Phi(s)$. Beside vertical propagation, if sectors share upstream suppliers or downstream buyers, then independent, idiosyncratic sectoral shocks generate non-vanishing aggregate GDP fluctuations:

$\liminf_{m \rightarrow \infty} \text{var}(Y) > 0$. As many horizontally-related sectors move in response, an idiosyncratic shock does not average out, and aggregate volatility survives even as the economy grows.

COROLLARY 3 (Granular sectoral weights)

Economically large sectors amplify aggregate fluctuations induced by horizontal network propagation.

► Go back

THEOREMS 2 AND 3: HORIZONTAL WEIGHTING AND IMPORTANCE

A

THEOREM 2 (On the weighting of network “economic” distances)

In an Input-Output economy, the horizontal dimension of a network (defined by demand- and supply-based distance matrices) is not weighted by its vertical dimension (defined by the directed or Leontief inverse matrix).

THEOREM 3 (On the importance of network “economic” distances)

Consider a network of size- m with Input-Output matrix $\mathbf{H}_m := \mathbf{V}_m \cup \mathbf{D}_m$, where $\mathbf{V}_m$ encodes vertical buyer-supplier ties and $\mathbf{D}_m$ encoding horizontal complementarities. Let $\boldsymbol{\epsilon}_m$ be a vector of independent shocks, and let $\mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m$ and $\mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m$ denote the associated matrices for propagation channels. Then, as the network size grows, the vertical propagation vanishes in spectral norm:

$$\left\| \mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m \right\|_{\Delta} / \left\| \mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m \right\|_{\Delta} \xrightarrow{m \rightarrow \infty} 0.$$

COROLLARY 2 (Endogenous expansion)

In growing networks, the horizontal geometry asymptotically dominates shock propagation even as vertical chains expand.

► market share, after redefinitions Input-Output tables

- square matrix, $\mathbf{X}_{m \times m}$, whose diagonal correspond to sector-share of own production while:
 - ★ $\mathbf{X}_{m \times m}^{up} = uppertriangular(\mathbf{X})$ is the share of production of sector-s that is bought by sector-s';
 - ★ $\mathbf{X}_{m \times m}^{dw} = lowertriangular(\mathbf{X})$ is the share of production that sector-s buys from sector-s'.
- in the dataset built, I compute $\mathbf{x} = \mathbf{1} - diag(\mathbf{X})$ to extract the *total share* of production that sectors buy from other sectors, where $\mathbf{1}$ an $m \times 1$ vector of ones

► commodity-by-commodity total direct requirements Input-Output tables

- square matrix, $\mathbf{H}_{m \times m}$, in which each entry corresponds to:
 - ★ $\mathbf{H}_{m \times m}^{up} = uppertriangular(\mathbf{V})$ is the dollar value of one unit of sector-s that is bought by sector-s';
 - ★ $\mathbf{H}_{m \times m}^{dw} = lowertriangular(\mathbf{V})$ is the dollar value of sector-s in order to buy one unit from sector-s'

→ *row sectors identify supplier (upstream), and column sectors identify buyer (downstream)*

INTERPRETING AN INPUT-OUTPUT MATRIX, CELLS

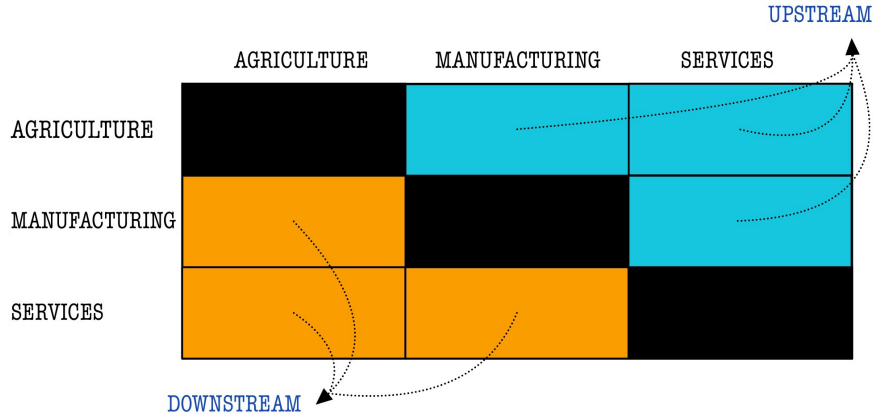
B - 2/5

	AGRICULTURE	MANUFACTURING	SERVICES
AGRICULTURE		AGRICULTURE SELLS TO MANUFACTURING	AGRICULTURE SELLS TO SERVICES
MANUFACTURING	AGRICULTURE BUYS FROM MANUFACTURING		MANUFACTURING SELLS TO SERVICES
SERVICES	AGRICULTURE BUYS FROM SERVICES	MANUFACTURING BUYS FROM SERVICES	

► Go back

INTERPRETING AN INPUT-OUTPUT MATRIX, TYPES OF SECTORS

B - 3/5



► Go back

- ▶ To investigate the network structure:
 - upstream and downstream sectors, as well as sectoral distances, are computed through the *commodity-by-commodity direct requirements* table as of 2007;
 - only to compute distances, disregard small transactions

$$\alpha^j(s, s') = \frac{\alpha(s, s')}{\mathcal{K}^j(s)} \text{ is set to } 0 \text{ if } \alpha^j(s, s') \leq 0.01$$

where $\mathcal{K}^{fd}(s) = \sum_{s' \in row} \alpha(s, s')$ and $\mathcal{K}^{fs}(s) = \sum_{s' \in column} \alpha(s, s')$

- ▶ Defining $\mathcal{K}^j = [\mathcal{K}^j(s), \mathcal{K}^j(s'), \dots, \mathcal{K}^j(m)]'$, the considered Input-Output structure is

$$\mathbf{H}_{m \times m}^j = \mathbf{H}_{m \times m}' \mathbf{K}_{m \times 1}^j \quad \text{for } j = \{fd, fs\}$$

in which each cell, $\alpha^j(s, s')$, is greater than 1%

- Exposure to inter-sectoral trade can be twofold:

① *direct*, which accounts for the exposure of sector- s to its supply chain;

→ however, also a certain intermediate input is the outcome of a separated supply chain. As a result, since sector- s buys intermediate inputs from other sectors, it is exposed also to the supply chain of each sector $s' \neq s$. Therefore ...

② *indirect*, which accounts for the exposure of sector- s to its supply chain, and to that of its trading-partner sectors. For instance, all intermediate inputs of sector- s , $X(s)$, are given by:

$$\begin{aligned} x(s) &= \underbrace{\alpha(s,s) x(s,s) + \alpha(s,s') x(s,s') + \dots + \alpha(s,m) x(s,m)}_{\text{direct}} \\ &= \underbrace{\alpha(s,s) x(s,s)}_{\text{direct}} + \underbrace{\alpha(s,s') \alpha(s',s) x(s,s) + \alpha(s,s') \alpha(s',s') x(s',s') + \dots + \alpha(s,s') \alpha(s',m) x(m,m)}_{\text{indirect, } s'} \\ &\quad + \dots + \underbrace{\alpha(s,m) \alpha(m,s) x(s,s) + \alpha(s,m) \alpha(s,s') x(s',s') + \dots + \alpha(s,m) \alpha(m,m) x(m,m)}_{\text{indirect, } m} \end{aligned}$$

→ which, in matrix form, is $\mathbf{X}_{m \times 1} = \mathbf{H}'_{m \times m} \mathbf{X}_{S \times 1}$, so to have that $\underbrace{\begin{pmatrix} \mathbf{I}_{m \times m} & -\mathbf{H}_{m \times m} \end{pmatrix}^{-1}}_{\mathcal{H}, \text{ Leontief inverse of matrix } \mathbf{H}} \mathbf{X}_{m \times 1} = \mathbf{0}$

NETWORK REPRESENTATION

share of sectoral total sales

B

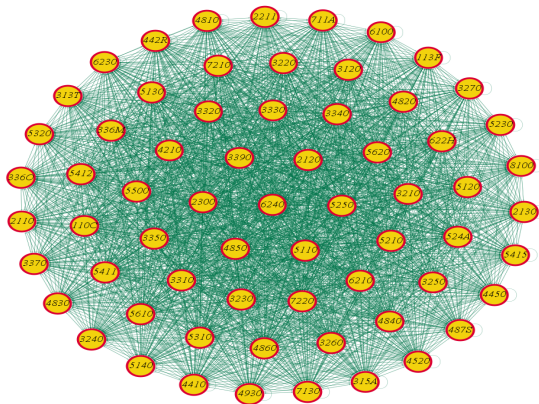


Figure B.1: Production network of the U.S. economy in 2007

► Go back

NETWORK REPRESENTATION

share of sectoral total purchases

B

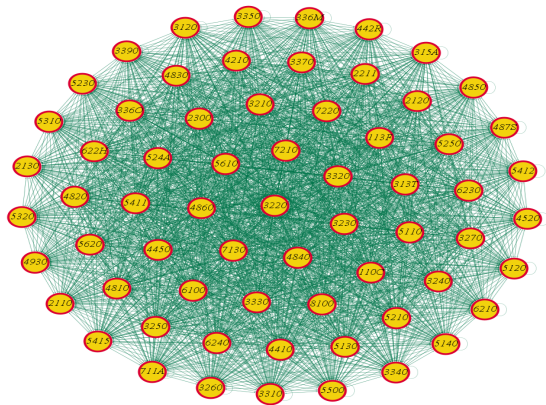


Figure B.2: Production network of the U.S. economy in 2007

► Go back

- System of *Cartesian coordinates* to compute the *network distances* {e.g., [Conley and Dupor \(2003\)](#)}:

- ① *factor input demand*, when buying similar inputs. Given $\alpha^{fd}(s, s') = \frac{\alpha(s, s')}{\sum_{s' \in \text{row}} \alpha(s, s')}$, then

$$d^{fd}[s, s'] = \left\{ \sum_{s^* \in \Phi(s)} \left[\alpha^{fd}(s, s^*) - \alpha^{fd}(s', s^*) \right]^2 \right\}^{\frac{1}{2}}$$

is measuring downstream proximity of production function requirements.

► Heatmap, demand

- ② *factor input supply*, when selling to similar sectors. Given $\alpha^{fs}(s, s') = \frac{\alpha(s, s')}{\sum_{s' \in \text{column}} \alpha(s, s')}$, then

$$d^{fs}[s, s'] = \left\{ \sum_{s^* \in \Phi(s)} \left[\alpha^{fs}(s^*, s) - \alpha^{fs}(s^*, s') \right]^2 \right\}^{\frac{1}{2}}$$

is measuring upstream comovement of sectoral revenues from trade;

► Heatmap, supply

- Distances are computed as those from the *directed network*

► Distributions

► Correlation

- the Leontief inverse transformation, instead, amplifies the magnitude of interconnections

SECTORAL DISTANCE IN THE NETWORK, FACTOR INPUT DEMAND

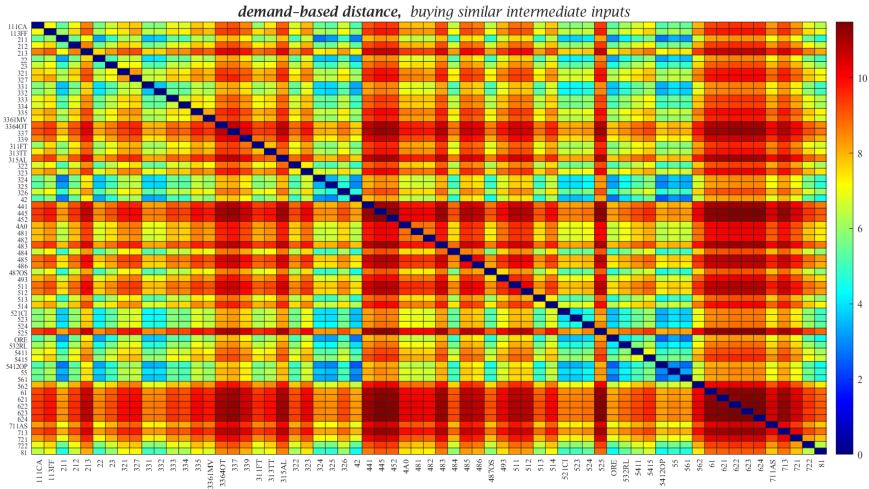
 $B - 1/2$ 

Figure B.3: *Factor demand distances in the Input-Output matrix, U.S. 2007*

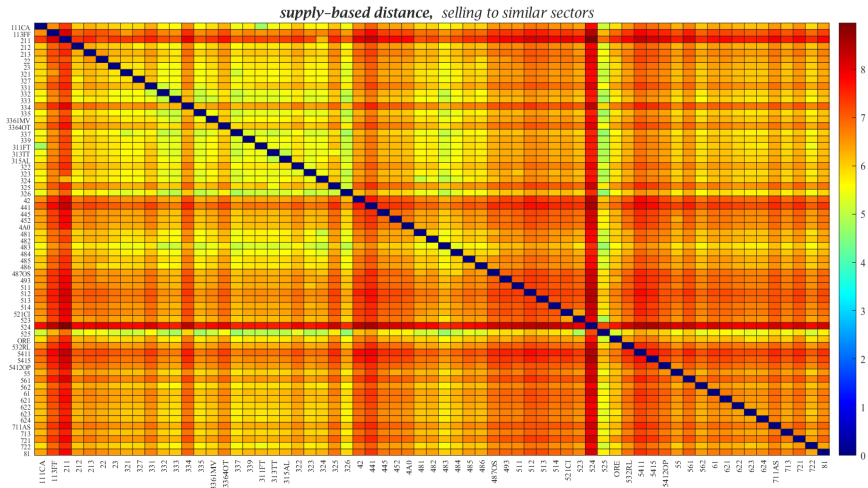
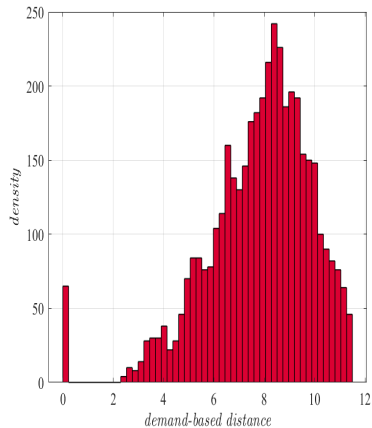
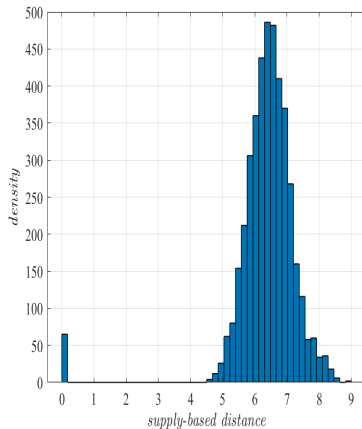


Figure B.4: Factor supply distances in the Input-Output matrix, U.S. 2007



(a) Factor input demand



(b) Factor input supply

Figure B.5: Distribution of network economic distances

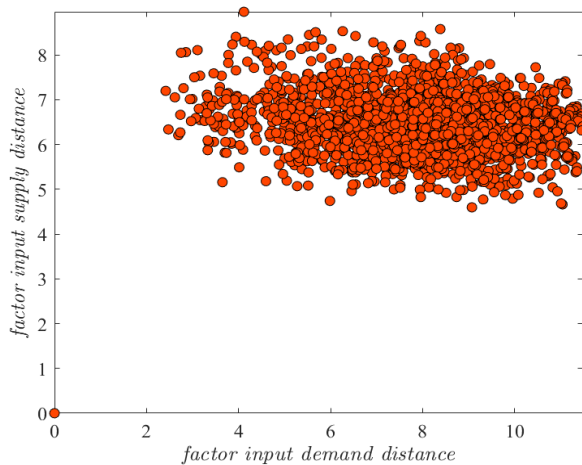


Figure B.6: Correlation of network economic distances

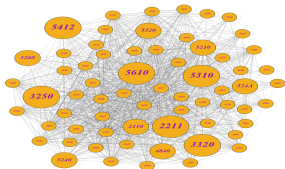
Shortest path algorithm.

Central idea is to find, for each single node, all the other nodes that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there – this is the shortest path approach. In other words, find the shortest number of steps it takes to get from every node to every other node in a network. In particular:

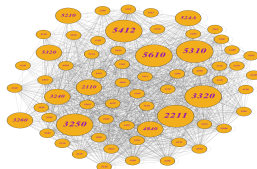
- (a) identify the direct connections between any pair of nodes (those that take exactly one step to be connected);
- (b) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.).

Multiply the graph by itself to discover which new pairs of nodes are now connected through longer paths;

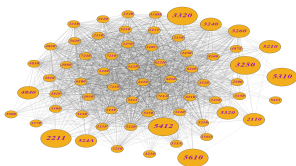
- (c) if a pair of nodes becomes connected (*i.e.*, only if it has not be already found a shorter path), record the current length as the shortest distance between them;
- (d) repeat this process until any new reachable pairs of nodes is found. For any pair of nodes that are still unreachable (*i.e.*, the distance is still zero and they are not linked in the network), set their distance to infinity.



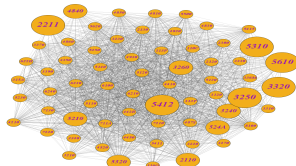
(a) Demand, $d = 1$



(b) Supply, $d = 1$



(c) Demand, $d = 2$



(d) Supply, $d = 2$

Figure B.7: Visualization of production network distances

- (1) The distribution of horizontal complementarities is uneven
 - the factor input demand network is slightly denser than the supply network at $d = 1$, but then this asymmetry vanishes ($d = 2$)
 - (2) Sectors with the major Input-Output linkages tend to concentrate around the nucleus
 - the most connected hubs (top 10%) and the mid-hubs (top 20%) are similarly central
 - (3) Sectors with the highest number of production network connections tend to be evenly central at shorter distances, and relatively more peripheral at longer distances
- 👉 The structure of horizontal complementarities depend on the notion of distance
- the same vertical production system generates distinct horizontal geometries (Theorem 1)

- ▶ sectoral output and employment growth, $\dot{var}_t(s) = var_t(s) - var_{t-1}(s)$, $var = \{y, n\}$;
- ▶ network distance- d computed as a *shortest path problem*, unweighted (Theorem 2);
- ▶ given Φ_s^j the set of sectors which are $k \in \{upstream, downstream\}$ to sector- s :
 - (a) distance- d growth, $\dot{var}_t(\Phi_s, d) = \sum_{s' \neq s} [var_t(s', d) - var_{t-1}(s', d)]$;
 - (b) type- k growth, $\dot{var}_t(\Phi_s, k) = \sum_{s' \neq s} [\ell(s, s' | k) (var_t(s', k) - var_{t-1}(s', k))]$;

where $\ell(s, s')$ is the Leontief inverse weight of sector- s' on sector- s
- ▶ controls:
 - (a) sectoral employment size, $n(s)$, value-added $y(s)$, and their growth rates, and net exports;
 - (b) aggregate employment size N and value-added Y , and their growth rates.

① Purge the dynamics of sectoral I-O shares from movements in sectoral employment

- panel Fixed-Effects (FE) regression over $s \in \Phi(s)$ sectors

$$\begin{aligned}
 x_t(s) = & \underbrace{\beta_n \dot{n}_t(s)}_{\text{emp. growth}} + \underbrace{\sum_d \beta_{n(d)} \dot{n}_t(\Phi_s, d)}_{\text{emp. growth, network distance}} + \underbrace{\sum_k \beta_{n(k)} \dot{n}_t(\Phi_s, k)}_{\text{emp. growth, upstream and downstream}} \\
 & + \underbrace{\beta_{z(s)} Z_t(s) + \beta_{\dot{z}(s)} \dot{Z}_t(s)}_{\text{sector-level controls}} + \underbrace{\beta_z Z_t + \beta_{\dot{z}} \dot{Z}_t}_{\text{aggregate controls}} + \underbrace{\psi(s)}_{\text{sector Fe}} + u_t^x(s)
 \end{aligned}$$

- estimated residuals $(\hat{u}_t^x(s))$ identify the residual variation in I-O shares

- ② Estimate panel Local Projections (LP) to study the dynamic effects of changes in sectoral I-O intermediate shares on its (cumulative) employment

- given $\bar{h} = 5$ horizons, sequence of panel predictive regressions of the form

$$\dot{n}_{t+\bar{h}}^{cum}(s) = \beta_{\hat{u}^x, \bar{h}} \hat{u}_t^x(s) + \underbrace{\psi(t + \bar{h}) + \psi_{\bar{h}}(s)}_{\text{horizon and sector Fe}} + v_{t+\bar{h}}^x(s)$$

... also considering d^{th} -distance across sectors

$$\dot{n}_{t+\bar{h}}^{cum}(s) = \beta_{\hat{u}^x(d), \bar{h}} \sum_{s' \neq s} \hat{u}_t^x(s', d) + \underbrace{\psi(t + \bar{h}) + \psi_{\bar{h}}(s)}_{\text{horizon and sector Fe}} + v_{t+\bar{h}}^x(s, d)$$

- variables at $\bar{h}$ identifies the cumulative difference between time t and $t + \bar{h}$

NETWORK SHARES AND EMPLOYMENT

C - 1/2

factor input demand and directed network

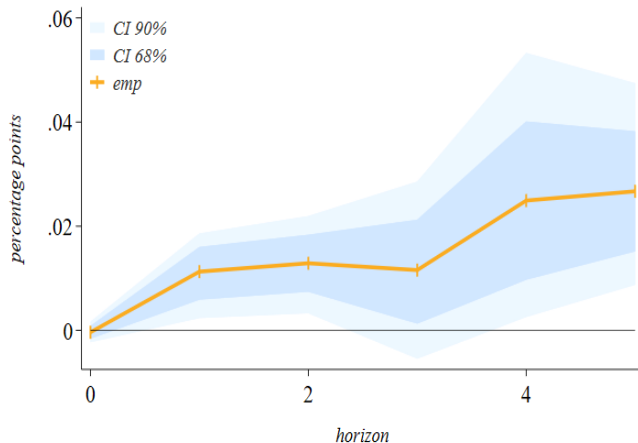
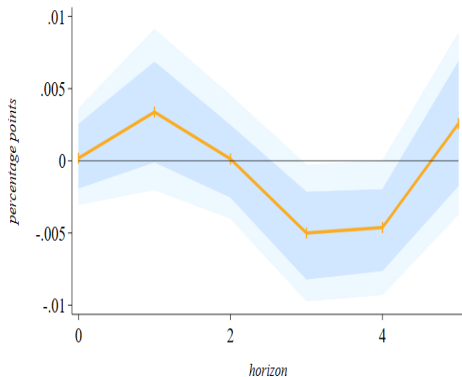


Figure D.1: Sectoral employment response following a 1% increase in own I-O share

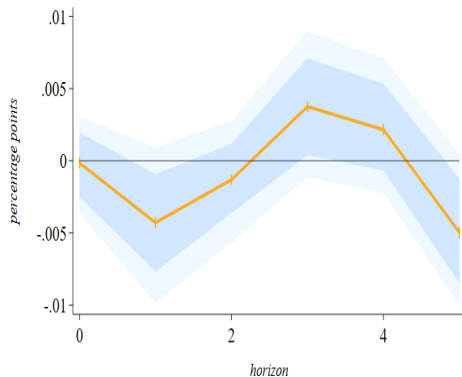
NETWORK SHARES AND EMPLOYMENT, DISTANCE

C - 2/2

factor input demand and directed network



(a) $d = 1$



(b) $d = 2$

Figure D.2: Sectoral employment response following a 1% increase in I-O share of nearby sectors

NETWORK SHARES AND EMPLOYMENT

factor input supply and directed network

C - 1/2

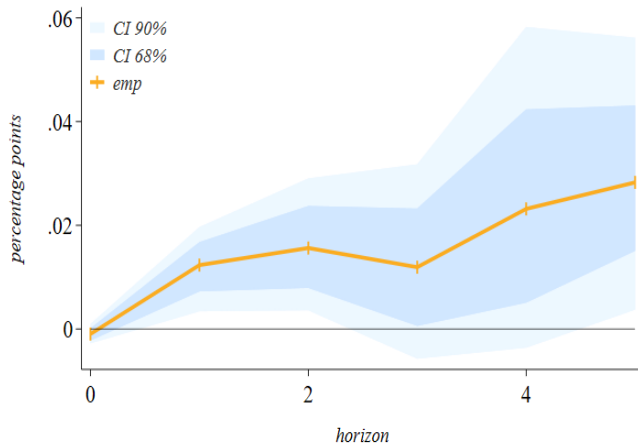
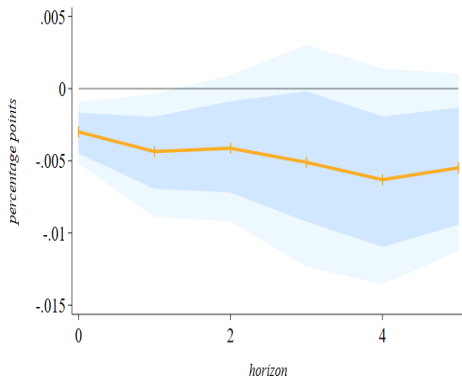


Figure D.3: Sectoral employment response following a 1% increase in own I-O share

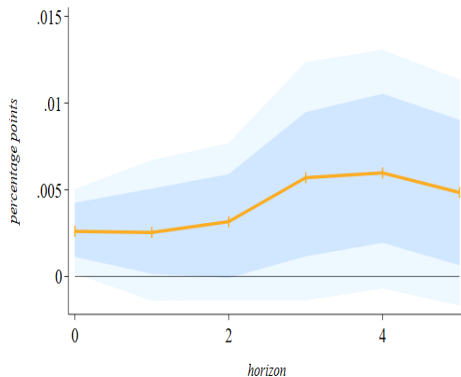
NETWORK SHARES AND EMPLOYMENT, DISTANCE

factor input supply and directed network

C - 2/2



(a) $d = 1$



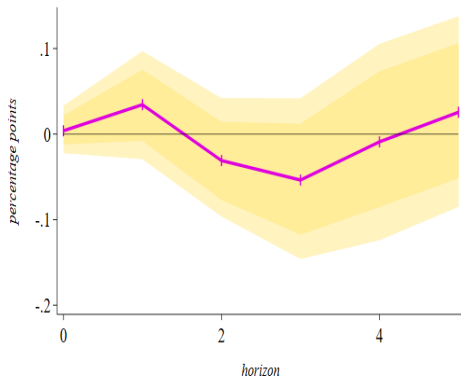
(b) $d = 2$

Figure D.4: Sectoral employment response following a 1% increase in I-O share of nearby sectors

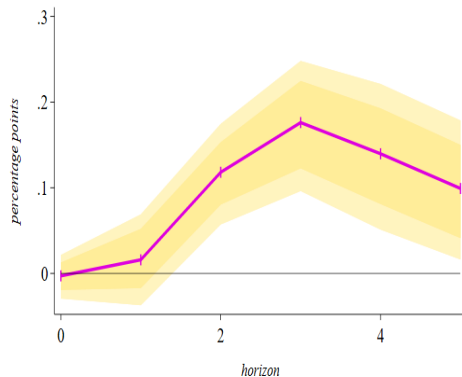
EMPLOYMENT EFFECTS, DISTANCE

factor input demand and directed network

C



(a) $d = 1$



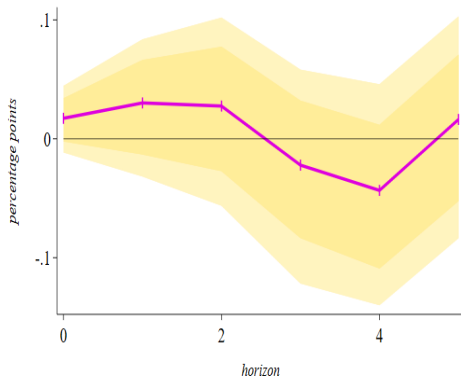
(b) $d = 2$

Figure E.1: Employment comovement under direct demand linkages

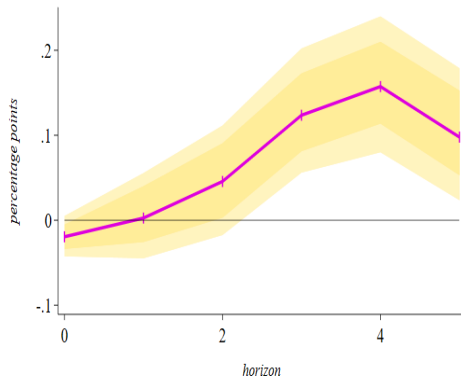
EMPLOYMENT EFFECTS, DISTANCE

factor input supply and directed network

C



(a) $d = 1$



(b) $d = 2$

Figure E.2: Employment comovement under direct supply linkages